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High costs of capital hinder the equity, speed and abatement potential of solar and wind deployment globally

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Abstract:

High costs of capital are a substantial barrier to renewables deployment in developing countries, but there is limited empirical data on how financing costs vary across countries and between technologies. Here, using bespoke estimates for over 150 countries, we assess the impact of national costs of capital on renewable levelised costs of electricity (LCOE). We show that the cost of capital is inversely related to economic development, with the inclusion of national costs of capital increasing LCOE in developing countries by an average of US$19/MWh and 28/MWh for solar and onshore wind, compared to a uniform scenario. Across developing regions, the share of financing costs in renewable LCOEs rises from 40-42% up to 78% under national costs of capital, leading to an annual cost increase of US$574bn and US$888bn globally if 5% of technical potential was developed. Factoring in grid carbon intensity, we show that the largest lifetime abatement potentials are located in developing countries, up to 20 times those in developed countries. Understanding the influence of high costs of capital on renewable costs will be key to the effective provision of international support in an increasingly constrained development finance environment, contributing to a more rapid and equitable energy transition.

Introduction

Global investment in renewables needs to rise to US$1.5 trillion per year between 2024-2030[1], if the world is to achieve the COP28 goal of tripling global renewable capacity by 2030 to at least 11,000 GW[2]. Whilst annual investment into renewables and other clean energy technologies has risen rapidly over the recent decades, driven by falling costs and increased technological maturity, annual investment only reached US$623bn in 2023[3]. Flows of finance vary drastically by region, with China, Europe and the US accounting for nearly 80% of global investment into renewables in 2023[3], but only covering 59% of global electricity demand .

Tripling global renewable capacity by 2030 is regarded as a key step in achieving net zero emissions globally by 2050, supported by modelling conducted by the IEA[4,5] and IRENA[6] and committed to by nearly 200 countries at COP28[7]. Wind and solar photovoltaic (hereon solar) power are expected to meet the vast majority of the COP28 target by both capacity and generation[4,6]. Over recent decades, both technologies have experienced notable cost reductions, exceeding most expert forecasts and predictions[8,9]. Renewable electricity is increasingly cost competitive with fossil fuel power on a levelised cost of electricity basis[10,11]. Despite their increasing cost competitiveness, the world is still on track to fall 30% short of the COP28 goal under national deployment commitments[12]. Strategic policy support, further cost reductions, and effective mobilisation of private capital into developing countries will be essential to accelerating progress toward the COP28 goal[13].

Renewable cost reductions over recent decades have been driven by four main factors: (1) Capital costs falling with experience as wind and solar have progressed along their learning curves[14]. (2) Decreasing cost of capital, originating from financial learning[15] and periods of low interest rates post-2008[16]. (3) Technical innovations that have increased the average capacity factor[17]. (4) Increased efficiency in operations[18]. However, whilst there have been drastic global cost reductions, the extent to which these have led through to reductions in regional levelized costs and ensuing deployment has been varied[11].

In developing countries, factors such as skill shortages[19], fossil fuel subsidies[20], limited grid integration capacities[21], and inadequate policy and regulatory setups[22] combine to hinder renewables deployment. However, a principal barrier is insufficient flows of energy investment at reasonable costs of capital[23–25]. Of the $1.3trn of climate finance mobilised in 2023, just over 2% went to the least developed countries, rising to 14% for all developing countries excluding China[26]. From an investor perspective, there are a distinct lack of “bankable” projects, where the risk-return profile fits the investor’s risk appetite[27]. Risk appetite differs between actor categories, with public financiers more likely to take on higher risks[28]. For private actors, higher returns (and so higher costs of capital for project developers) are required for investments with higher risk perceptions[29], with significant risk premiums demanded for projects in developing countries[30].

The costs of capital from private investors (referred to onwards as commercial capital) for energy investments is therefore over twice as high in developing countries than in the developed world, driven by a range of risk factors at the project, sector and country level[31]. Many of these risks are macroeconomic in nature and so are outside of the control of energy policymakers[31], including currency risks[32], political instability[33] and the strength of state institutions[34]. These risks lead to developing countries being stuck in a “climate investment

trap", as described by Ameli.[35], where climate-related investments remain chronically insufficient through self-reinforcing mechanisms. The impacts of high costs of capital on developing country energy transition are compounded by the significant role that finance plays in determining the cost of renewable electricity[36], due to high capital intensity and negligible operating costs, with financing costs making up 40% of the LCOE even at low costs of capital[37]. The economic viability of renewables is therefore highly sensitive to financing costs[38–40], adding an additional barrier to the clean energy transition in developing countries[41].

High costs of commercial capital for clean energy technologies in developing economies have led to public and concessional financing (the latter defined as financing offered below market rates [42]) playing a larger role in energy investment. In 2022, public and concessional financing accounted for 58% of investment in developing economies (52% and 6%, respectively), compared to under 20% in advanced economies[43]. Despite its low share, concessional financing plays an important role in de-risking investments in developing countries. An estimated US$200-300bn of concessional finance will be required per year to stay in line with global climate targets[44], including through grants, guarantees, technical assistance and loans with concessional terms[45], to reduce the cost of capital facing clean energy projects in developing countries.

Drastic differences between the quality of renewable energy resources and accessible costs of capital also means that renewables deployment has historically been in locations with low costs of capital, rather than those that have the best renewable potential[46]. For example, at the time of writing the UK has more installed solar PV by capacity than the whole continent of Africa[47], despite vastly poorer incident solar resources[46]. The highest quality renewable energy potentials are located in areas that include North and South Africa & the Middle East (solar)[48] and Africa, Russia and Central Asia (wind)[49], which face higher costs of capital than countries and regions that account for the majority of deployment (the U.S., China and the European Union). Concessional financing and other de-risking programmes are therefore essential to support mobilisation of private capital at affordable rates into developing countries to support renewable deployment[50], but also to ensuring the decarbonisation of the global energy system is more efficient both in terms of energy generation[51] and carbon abatement per MW of installed renewables[52].

Existing work has highlighted the challenge high costs of capital poses to the energy transition in developing countries[24, 35,25]. Detailed exploration of this challenge is hampered by poor data availability, with terms of finance regarded as privileged information in a competitive environment[53]. Cost of capital estimation methods - with sectoral, national and temporal granularity - is a key area of interest for energy economists and modellers[54–57], as conclusions drawn from technoeconomic and energy system models are highly sensitive to cost of capital assumptions[36,39,58]. Developing estimates of the likely ranges of costs of capital in developing countries and the extent to which they constrain technical resource potential is therefore important to ensuring the accuracy and utility of technical insights[36,57], improving the relevance of modelled decarbonisation pathways[59].

Here, we present global estimates of the levelised cost of electricity (LCOE), annual technical potential and lifetime abatement of solar, onshore wind and offshore wind power at a high geospatial granularity, using country- and technology-specific costs of capital for over 150

countries in 2024. Combining the technical and economic analysis, we evaluate how much cost of capital raises the LCOE across over 96,000 grid points and the share that financing costs play in the overall LCOE. Regional cost-supply curves are generated, with quantification of the extent to which high costs of capital constrains renewable technical potential. Using national electricity grid carbon intensities, we also evaluate renewable abatement potential at a grid point scale and assess their cost competitiveness in abating existing fossil generation by region. We find that locations in Africa, Central Asia and South America would have levelized costs within or below US LCOE ranges, if countries could access developed countries' financing terms, with the share of financing costs in the overall LCOE rising from 40% to nearly 76%, an average worldwide increase of US$15.4/MWh (solar) and US$37.8/MWh (onshore wind).

Modelling the Cost and Scale of Renewable Potential

Our modelling approach entails five distinct steps, demonstrated by the schematic in **Figure 1**. First, we develop hourly power output profiles of solar PV[60] and wind power[61] globally. Secondly, we create estimates of country- and technology- costs of capital for 176 countries for 2024, adapting an existing, peer-reviewed approach to widen its scope and geographical coverage (see **Methods**)[54]. Thirdly, we calculate the LCOE based on national or regional deployment costs for solar and wind power, with estimates of the cost of capital used for each country and technology, with the share of the LCOE attributable to the financing costs evaluated. We also perform the calculation based on a uniform cost of capital scenario, using the average estimated cost of capital in the UNFCCC Annex II countries for the relevant technology (hereon referred to as "developed" countries). Fourthly, we estimate the technical renewable potential (i.e., the maximum electricity production per km^2 per year) at each grid point using land use availability factors and packing density limits for each technology, which is then aggregated to a national and regional level. Finally, the renewable abatement potential at each location is also calculated – using technical potential, renewable lifetime and grid intensity data – which is then used to generate LCOE-abatement curves.

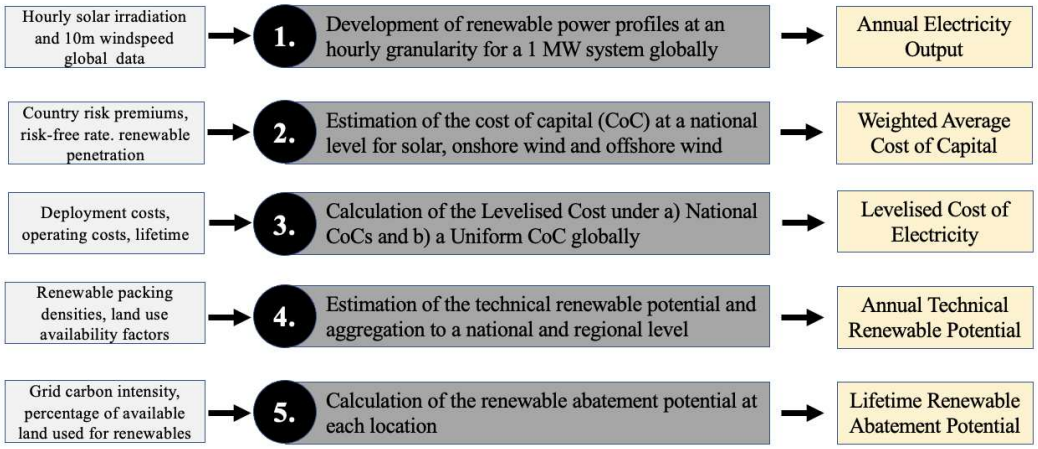


Figure 1: A schematic of the modelling approach used in this work. *The five modelling steps used in this work are outlined, with inputs highlighted in light grey on the left and the main outputs from each step detailed in tan on the right.*

Global Renewable Energy Potentials

Figure 2 presents the annual solar, onshore wind and offshore wind technical potentials globally, obtained by applying our model with minimum capacity factor cut offs (see **Methods**). Potentials are ordered by estimated national cost of capitals, with the GDP per capita in each country[62] highlighted. The estimated cost of capital varies drastically across countries (see **ED Figure 2** for regional breakdowns), from 6.6 – 30.4%, 6.8 – 30.8% and 7.6 – 31.7% for solar, onshore and offshore wind in nominal terms (see **Methods**). Commercial financing terms for renewable energy in developing countries is up to triple the average in developed countries, which is 8.2%, 8.1% and 9% for solar, onshore and offshore wind (see **Supplementary** and **ED Figure 2** for data and geographic distribution of national costs of capital). Global annual potential is 830,000, 935,000 and 15,000 TWh per annum for solar, onshore and offshore wind, respectively, broadly in line with other estimates from the literature (see the comparison of global solar and wind potentials made by Chu[63]).

Despite substantial technical potential, in many cases higher quality than developed countries on a capacity factor basis, the majority of developing countries have costs of capital that are at least double those in developing countries (see **ED Figure 3** for the regional breakdown). The least developed countries on a GDP per capita face some of the highest costs of commercial capital - with an observable relationship between cost of capital and GDP per capita - which acts as a barrier to cost-effective renewables deployment and necessitates support from international financiers. It also risks further lock in of fossil fuel infrastructure in developing countries[64], holding implications for both the equity and success of the global energy transition without international support[24,25,45].

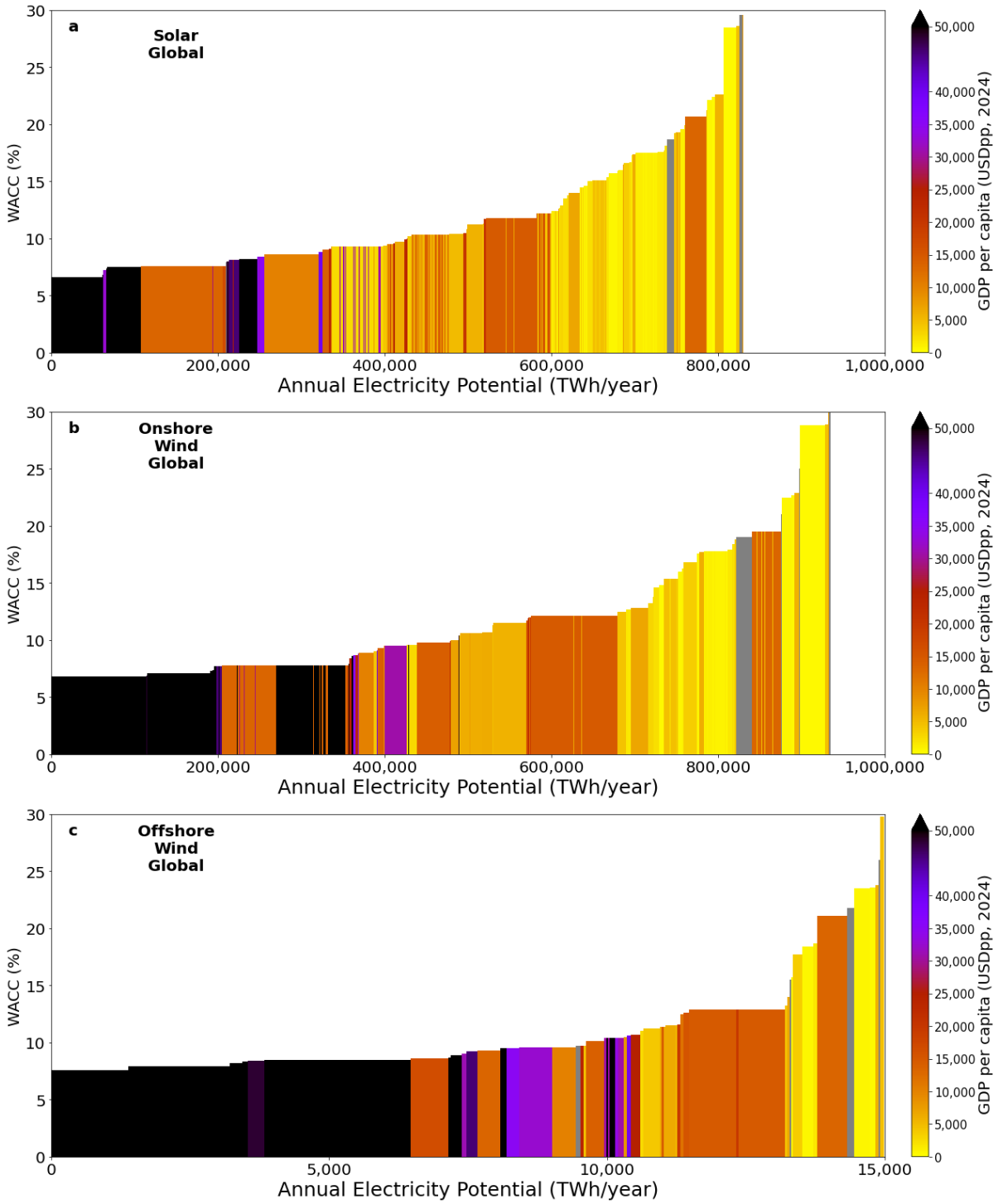


Figure 2: *Global technical potential curves for a) solar PV, b) onshore wind and c) offshore wind, ordered by estimated weighted average cost of capital for 2024 (see Methods). Note the different scale used for offshore wind potential. Potentials are aggregated at a country level, with shading indicating the level of economic development measured using GDP per capita (in PPP terms). Packing density and land use constraints were applied (see Methods). Minimum cut-offs of 10% and 18% average capacity factor were applied to the solar and wind potentials to reflect constraints in economic feasibility[63].*

High costs of capital hold back cost-competitive renewable potential

The levelised cost of electricity (LCOE) is one of the measures used to assess the cost competitiveness of different electricity generation projects and technologies[11,65]. Using an economic model with regionally and nationally specific deployment costs, the levelised cost of electricity for solar, offshore wind and onshore wind is evaluated with a high spatial granularity under two cost of capital scenarios: 1) National, using technology and country-specific cost of capital, and 2) Uniform, where the average cost of capital in nominal terms in developed countries for the relevant technology is applied globally. **Error! Reference source not found.** shows the change in the levelised cost of electricity from each technology between the National and Uniform scenario (spatial plots of LCOE under the two scenarios included in **ED Figure 5** and **ED Figure 4)**. The average increase from national costs of capital is US$19 and 28/MWh for solar PV and onshore wind, respectively, an increase of 28 and 35%. For offshore wind, which involves larger capital expenditures, this rises to US$31/MWh (24%).

Total increases in cost of capital for all regions account to US$574 and US$888bn for onshore wind and solar PV, respectively, if 5% of all technical potential was developed. Increases in LCOE from the Uniform to National scenario are highest in countries in Africa, Central Asia and parts of South America, exceeding US$150/MWh for wind and US$100/MWh for solar, whilst in Western Europe and North America there is little or no change in the LCOE (with national costs of capital close or equal to the Annex II average). Countries currently in military, political or macroeconomic crises (e.g., Ukraine, Venezuela, South Sudan) see the largest increases in LCOE. Assessment of the breakdown of their costs of capital show that these increases are driven almost exclusively by the country risk premiums, a financial measure of the perceived risk involved with investing in a given country[34].

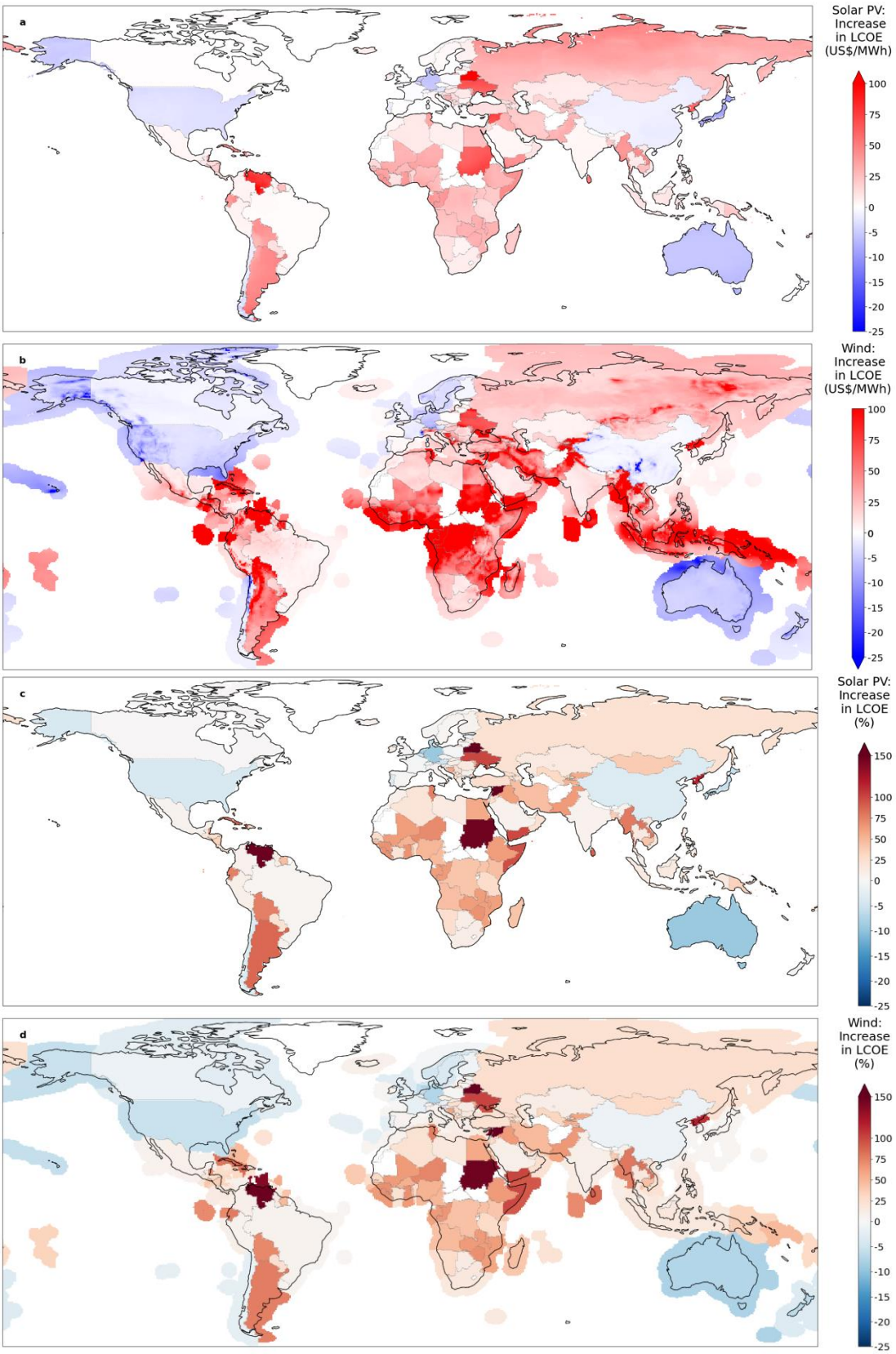


Figure 3: Change in the levelised cost of electricity of a) solar PV, b) onshore and offshore wind from the Uniform to National Scenario, in nominal terms. Percentages changes in LCOE are plotted in c) for solar PV and d) for wind power. *The National Scenario uses country and technology specific estimates of the cost of capital, whilst the Uniform Scenario uses a uniform cost of capital of 8.2%, 8.1% and 9.0% for solar, onshore wind and offshore wind, respectively, which is the average cost of capital in the UNFCCC Annex II countries.*

Financing costs already make up a substantial portion of the overall LCOE for solar and wind in developed countries[37], due to their high capital intensities and low operational costs[39]. **Figure 4** plots the share of the LCOE attributable to financing costs at each grid cell modelled, which ranges from 36-78% across technologies and regions in the National scenario. In the Uniform scenario, by contrast, financing costs contribute an average of 41-42% to the LCOE across technologies. In line with the distribution of cost of capital, the lowest shares of finance to the LCOE under the National scenario are in developed countries, with high costs of capital substantially pushing up the LCOE in developing countries[39]. Both results highlight that the LCOE is extremely sensitive to the cost of capital, as found in other studies[38–40].

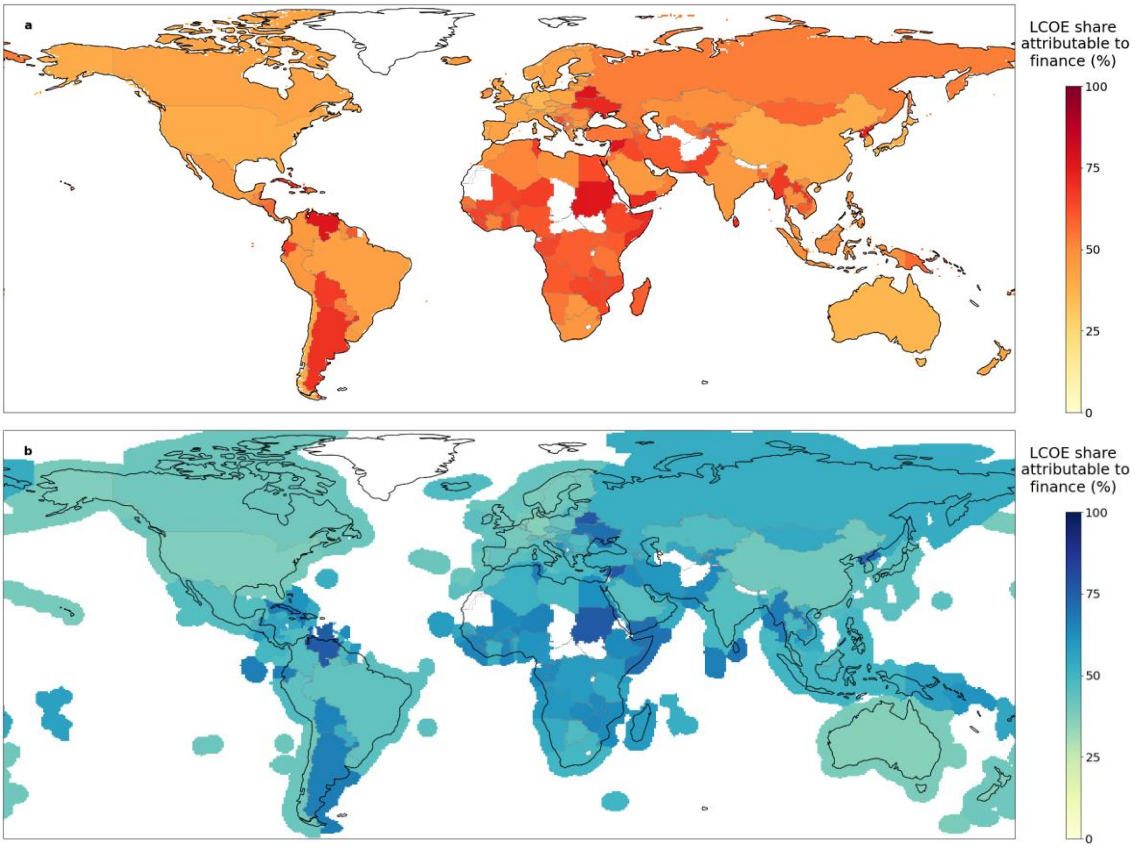


Figure 4: The share of the levelised cost of electricity (LCOE) attributable to financing costs, mapped globally for a) solar PV and b) wind power*. Results are depicted for the Country scenario, which uses country- and technology- specific costs of capital to calculate the LCOE. The share of financing costs is calculated by subtracting the contributions of capital costs and operating costs to the overall LCOE and then evaluating it as a proportion of overall LCOE.*

Understanding the extent to which low-cost renewable power can be scaled up, both nationally and regionally, is also important to long term energy planning and to the accurate design of policy and regulatory support[66]. **Figure 5** plots the regional LCOE-supply curves under both the National and Uniform scenarios for each technology, with regions grouped using the regional mappings from the TIAM-Global model[67]. The largest increases in LCOE are in Africa, Latin America and the Former Soviet Union (FSU), reaching US$115 and over 150/ MWh for solar and onshore wind (for locations within the minimum capacity factor bounds). The lowest achievable LCOEs globally are in China, reaching US$30 and 26 /MWh for solar and

onshore wind, due to a combination of low costs of capital and the lowest national deployment costs on a per kW basis globally[11]. For offshore wind, LCOEs reach as low as US$60/MWh in Western Europe, in line with recent offshore wind auction bids[68].

Africa has the largest renewable potential for both solar PV and onshore wind, reaching over 165,000 and 225,000 TWh per annum (5.4 and 7.5 times the volume of electricity generated globally from all sources in 2023[4]). Renewable energy potential in Africa is severely limited by high costs of capital, with the percentage of solar potential under the midpoint of Lazard's LCOE benchmarks[10] (US$55/MWh) falling from 89.7% to 23.5% due to high costs of capital. For onshore wind, the share within the range reported by Lazard (US$27-73/MWh) falls from 48.3% to 5.5%. Other developing regions also see similar (drastic) falls in the share of potential within the Lazard LCOE benchmarks from the Uniform to National scenario, particular solar in the Middle East (100% to 35%), solar in Other Asia (40% to 0.1%) and onshore wind in Central and South America (73% to 35%).

Historically, the majority of offshore wind development has taken place in China and European countries bordering the North Sea[69], reaching a global installed capacity of 72 GW in 2023[4] (259 TWh per annum, using an average capacity factor of 41%[11]). Global offshore potential estimated in this work exceeds this by nearly 60 times, reaching 15,000 TWh. However, the majority of locations evaluated in this work have LCOEs far higher than the Lazard benchmarks[10]. (US$74-139/MWh) and IRENA's reported global weighted average (US$75/MWh)[11]. The largest potential within the range of Lazard's benchmarks (i.e. below US$106/MWh[10]) is in Western Europe, at 1050 TWh per annum. There is substantial offshore wind potential in Central & South America and the Former Soviet Union countries, which could reach below US$106/MWh if countries were able to access the same cost of capital as developed countries (780 and 650 TWh per annum respectively). Under national costs of capital, this falls to 295 and 86 TWh in each region (15 and 5% of regional offshore potential). The cost of capital is even more important to offshore wind project viability given the scale and capital intensity of projects, which explains why many offshore wind developers have paused projects due to recent rises in global interest rates[70].

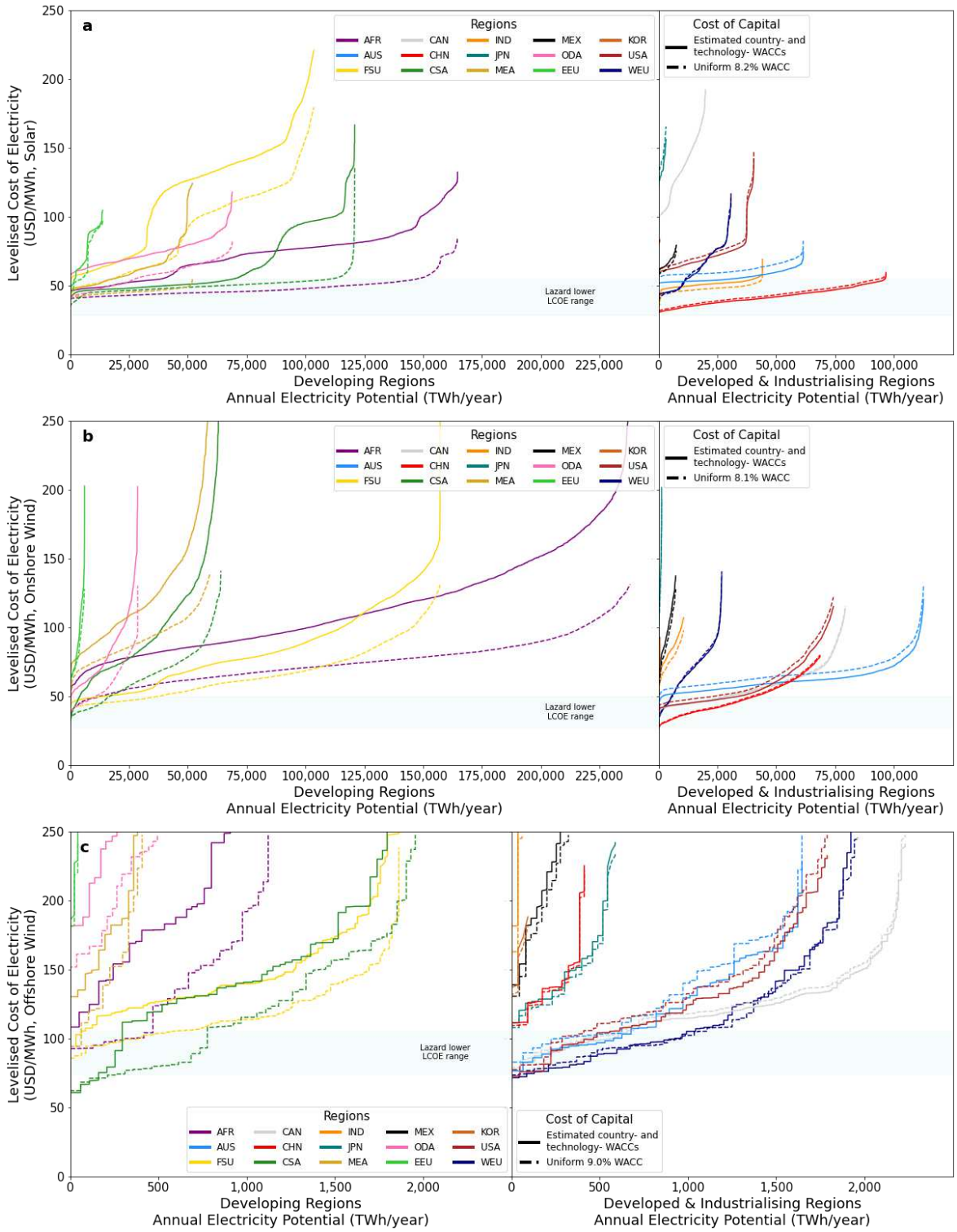


Figure 5: Regional cost-supply curves for a) solar PV, b) onshore wind and c) offshore wind, plotted for all 16 regions examined under the two cost of capital scenarios. *Note the different scale for electricity potential used for offshore wind. Levelised costs under the national estimates of technology cost of capitals are displayed using the solid lines, whilst the dotted lines depict the levelised cost in each region under the average technology cost of capital in the UNFCCC Annex II countries applied globally. AFR: Africa, AUS: Australia, FSU: Former Soviet Union, CAN: Canada, CHN: China, CSA: Central and South America, IND: India, JPN: Japan, MEA: Middle East, ODA: Other Developing Asia, EEU: Eastern Europe, KOR: Korea, USA: United States of America, WEU: Western Europe (see Supplementary for the full country-region mappings)*

Renewable abatement potential

High costs of capital have implications on both the equity and pace of the energy transition[24]. Deployment of renewables in locations with high grid carbon intensities and higher capacity factors will deliver larger emissions reductions per MW of installed capacity[52], though accounting for proposed future fossil fuel capacity additions may also shift the optimal locations for abatement[64]. As it is macroeconomic and market factors that drive the cost of commercial capital and subsequent investment decisions for renewables, there is typically weak links between the cost of capital for renewable investment, local renewable potential and national grid intensities. **Figure 6** plots the lifetime abatement potential for 1 MW of wind or solar capacity globally, based on current grid carbon intensities[47] (see **Methods**). Locations with the highest abatement potential include South Africa, Libya, Turkmenistan and Uzbekistan, over 20 times the abatement potential in developed countries e.g., France, Norway, and the United Kingdom. Excluding Australia and Poland, which still have substantial coal generation[71], it is predominantly developing and industrialising countries that have the highest grid intensities (see **Extended Data Figure 6**). These countries are also predominantly the locations where high costs of capital hinders the deployment of renewables.

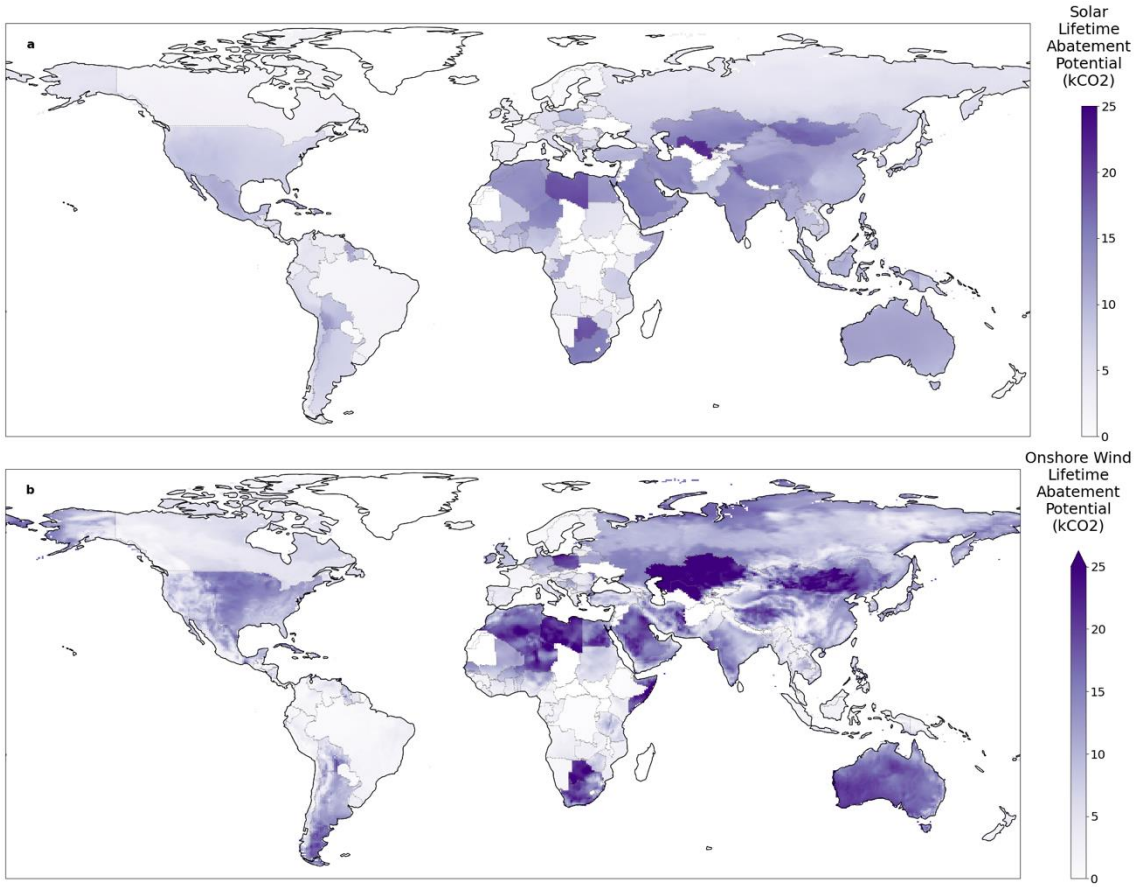


Figure 6: Carbon abatement potential for 1MW of installed a) solar PV and b) onshore wind, plotted globally in kilotonnes of CO_2 across the total project lifetime*. Calculations assuming that each kWh of renewable electricity generated displaces a kWh of grid electricity. Grid carbon intensity is taken from Ember for 2023 and is assumed to fall linearly to zero carbon emissions (see* ***Methods****), based on the need for power sector emissions to reach zero by 2045 to align with the Paris Agreement's targets[4].*

High costs of capital mean that the scales of abatement required in developing regions cannot be met at cost competitive LCOEs, for both solar and onshore wind. **Figure 7** plots the cumulative abatement potential against the levelised cost of electricity for solar and onshore wind at a regional level, with limits applied based on national annual electricity generation in 2024 (**see Methods**). The largest abatement potentials are in India, the U.S. and China, rising up to 14, 16 and 55 GtCO2, with the LCOEs within the Lazard benchmarks at the best locations. For developing regions, only in the Middle East and Latin America can a substantial portion of emissions be abated with solar at costs below the average Lazard benchmark of US$55/MWh, whilst for onshore wind it is only a negligible portion of abatement potential in Other Asia that is below US$50/MWh. Access to lower costs of capital observed for Other Asia, Africa (solar) and Former Soviet Union (wind), brings a larger portion of abatement potential within the Lazard benchmarks. However, even under the Uniform scenario a large portion of abatement potential is above these benchmarks.

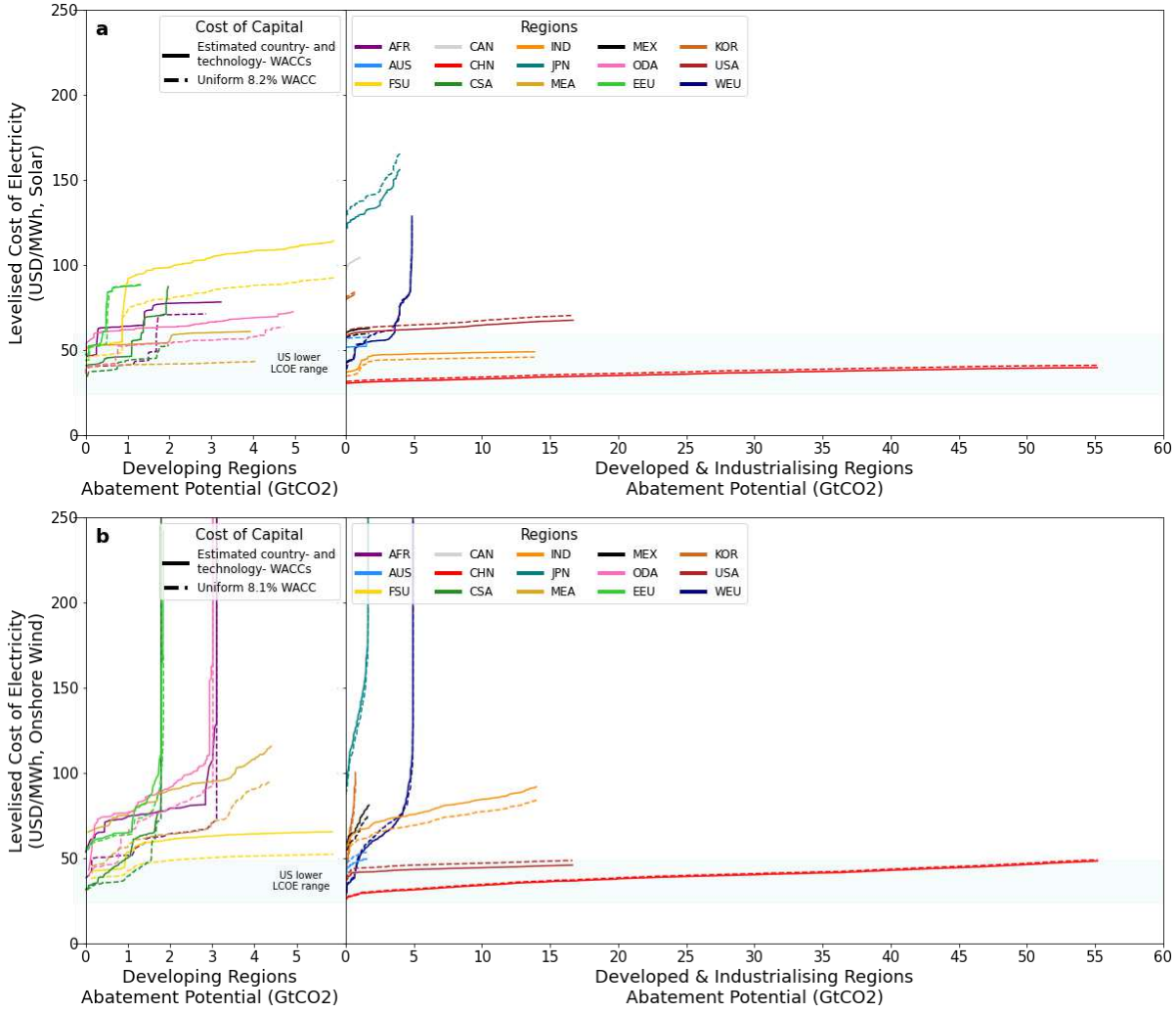


Figure 7: Lifetime abatement potential for a) solar PV and b) onshore wind, plotted at a regional level in gigatonnes of carbon dioxide against the levelised cost of electricity. *As with Figure 6, calculations assume that each kWh of renewable electricity generated displaces a kWh of grid electricity. Grid carbon intensity and total electricity generation is taken from Ember Climate[47], with the lowest cost locations assumed to be used to displace existing fossil generation (see **Methods**). AFR: Africa, AUS: Australia, FSU: Former Soviet Union, CAN: Canada, CHN: China, CSA: Central and South America, IND: India, JPN: Japan, MEA: Middle East, ODA: Other Developing Asia, EEU: Eastern Europe, KOR: Korea, USA: United States of America, WEU: Western Europe (see Supplementary for the full country-region mappings)*

Discussion

High costs of capital for renewable investments are a substantial barrier to the energy transition in developing countries. Addressing these barriers through policy interventions and concessional finance will be essential to an equitable and rapid energy transition[31]. Here, through a global technoeconomic evaluation of the potential, levelised cost and abatement potential of solar, onshore and offshore wind generation, we show where high costs of capital are holding back renewable deployment with high spatial granularity. A substantial portion of global high-quality renewable potential is held back by national costs of capital, causing an average increase in LCOE of US$15, 38 and 29/MWh (for solar PV, onshore and offshore wind, respectively). In developing countries that face the highest costs of capital, financing costs account for up to 78% of the share of the LCOE, nearly double the share if they could access the lower cost financing available in developed countries.

Our modelling demonstrates that the cost of capital for renewable energy investments is closely related to national GDP per capita, aligning with previous findings[72]. The least developed countries therefore face the highest costs of capital, over triple the average of developed economies. High costs of capital leads to inadequate levels of clean energy investment in developing countries[35], putting renewables at a competitive disadvantage and leaving the economies least able to fund their energy transition continuing to depend on fossil fuels and leading to subsequent vulnerability to price volatility[73]. For example, in Bangladesh, 62% of electricity demand growth has been met by expansion of gas power in the ten years to 2021. Rising LNG costs forced Bangladesh to halt purchases of LNG and implement load-shedding measures[74]. Expanding electricity supply through solar in Bangladesh could've saved US$2.7bn and reduced imports by 25%[75], but our modelling show that it faces a cost of capital nearly double that in developing countries (14%). Failure to bring down the cost of capital will mean that less developed countries, including Bangladesh, will be unable to access the economic benefits of falling renewable deployment costs [76].

Addressing the inadequate flows of energy investment to developing regions at reasonable costs of capital is not just motivated by a global energy justice perspective, though it is essential that this remains at the heart of the global response[77]. To avoid further lock-in of carbon-intensive infrastructure[73] and the subsequent risk of future asset stranding[78], ensuring that renewables are cost competitive in developing countries will be essential. Developing countries and regions are expected to account for high shares of the global growth in energy demand over the next few decades[31], which will need to be met by clean Estimates show that annual clean energy investment into developing countries will need to increase by more than six times by 2030 for the world to be on track for net zero by 2050[31].

Concessional financing will be important to effectively crowding in private investment required for the energy transition in developing countries, by serving as an anchor point and addressing underlying policy risk factors[79]. An estimated 70% of the risk factors contributing to high costs of capital in developing countries are caused by macroeconomic and currency risks[31], though policy risk also plays a key role[80]. Support from international actors proposed to address these risks include loan guarantee programmes[81], strengthening national regulatory frameworks[31], and provision of finance at concessional rates from international

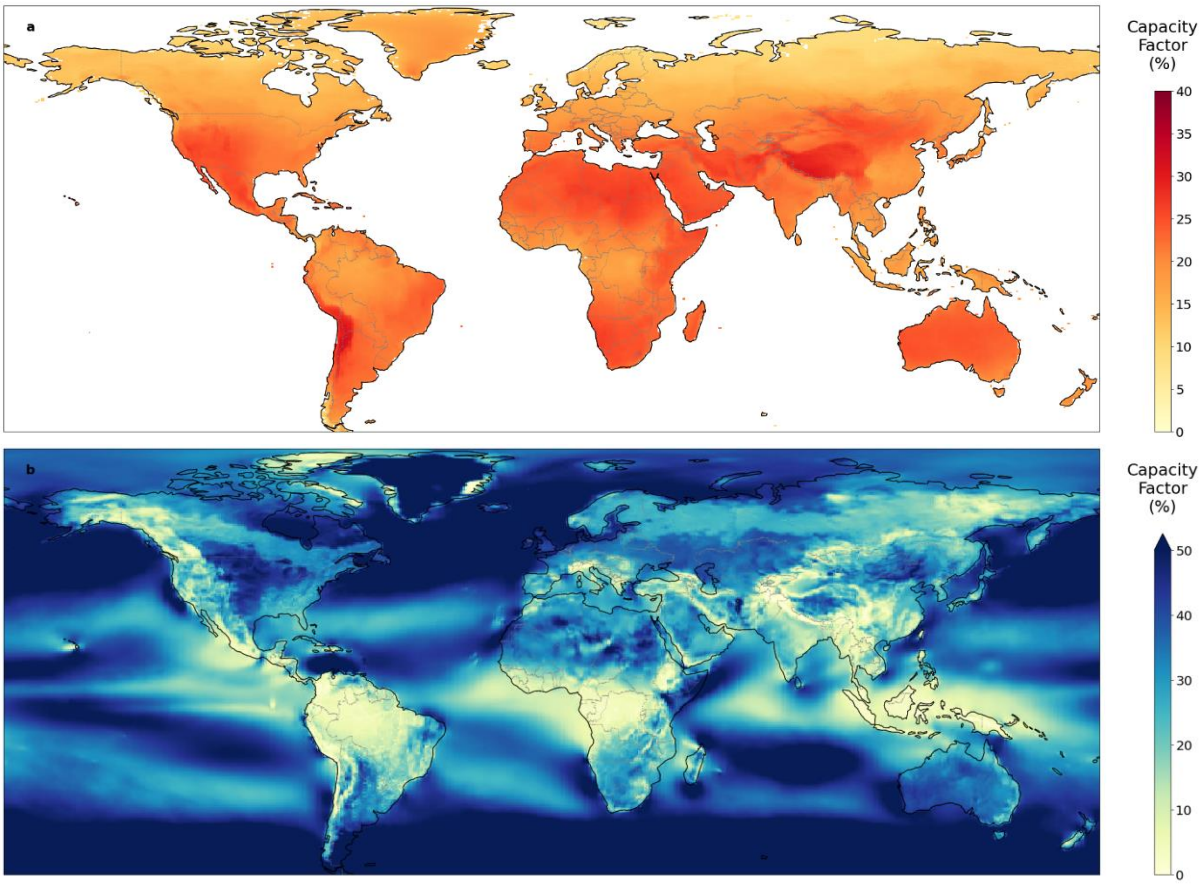
sources[25]. The latter is critical for the least developed countries, which we show face some of the highest commercial costs of capital. Estimates show that that concessional energy-related funding for clean energy investment will similarly need to scale up by five to six times to ensure that clean energy investment in developing countries reaches a level that delivers on the Paris Agreement[82,83]

In addition to direct reductions through concessional loans, international finance from public sources can also be used to support reductions in the cost of capital through measures such as capacity building and technical assistance[84]. The presence of international support can also attract private investors, who otherwise would not consider a specific investment "bankable". The New Collective Quantified Goal for climate finance, agreed at COP29[85], included an ambition to expand international public and concessional financing but was criticised for lacking sufficient inclusion of grants and other below-market sources of financing[86]. Here, we show that Sub-Saharan Africa, Latin America and parts of Asia see high costs of capital that severely limits renewables feasibility, and so international support in these regions would be most impactful to project viability.

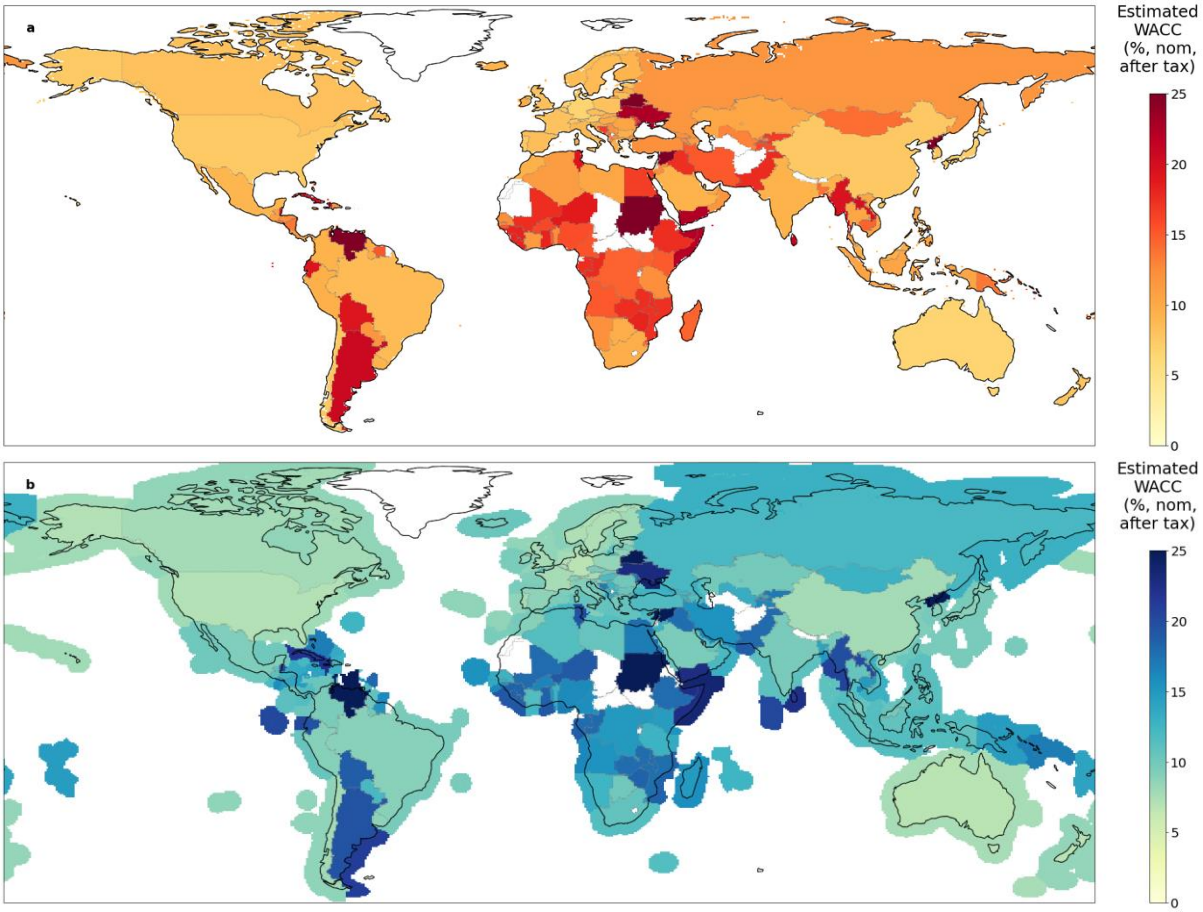
Whilst international flows of concessional climate finance will be important to achieving the COP28 goal of tripling renewables by 2030, domestic interventions such as market reforms[87], renewable energy subsidies[88] and coordinated national energy planning[89] are also required. Tailored support which accounts for national and regional circumstances will be important – for example, despite electricity costs from renewables falling rapidly over the last few decades[90], capital expenditures per kW of renewables capacity still range substantially across regions (up to a ratio of 3 in 2023)[11]. Revenue stabilization measures and use of auctions to award permits have been shown to de-risk renewable investments[91], whilst investment subsidies[88] could be used to improve short term viability in regions with high costs whilst the industry and workforce scales up to deliver economies of scale. Coordinating domestic policy measures and targets with the provision of international support, for example through Just Energy Transition Partnerships[92], will also play an important role.

Despite the key role that the cost of capital plays in the economics of low-carbon technologies[38], the regional approach used by many energy system models fails to capture the range of costs of capital within regions containing countries at disparate stages and contexts to the energy transition[93]. Inaccurate assumptions over the cost of capital have been shown to substantially bias model results[39], limiting the utility of energy research for policymakers. Here we develop estimates of the cost of capital, renewable potential, LCOE-supply curves and lifetime abatement at a high spatial granularity, which will serve as a valuable inputs to future technoeconomic[46] and energy system modelling efforts[72], addressing data availability issues highlighted by other works[56,94]. Using accurate, context-specific data is important to the usefulness of energy modelling, particularly in developing countries[95], and the results developing from this work will help support academics, policymakers and industry stakeholders to maximise the utility of technical insights from energy modelling tools. Further efforts should be focused on developing a robust methodology for projecting the cost of capital forward alongside broadening the coverage to emerging low-carbon technologies and accounting for subnational variance in the cost of capital for renewable firms[15] and projects[80,96].

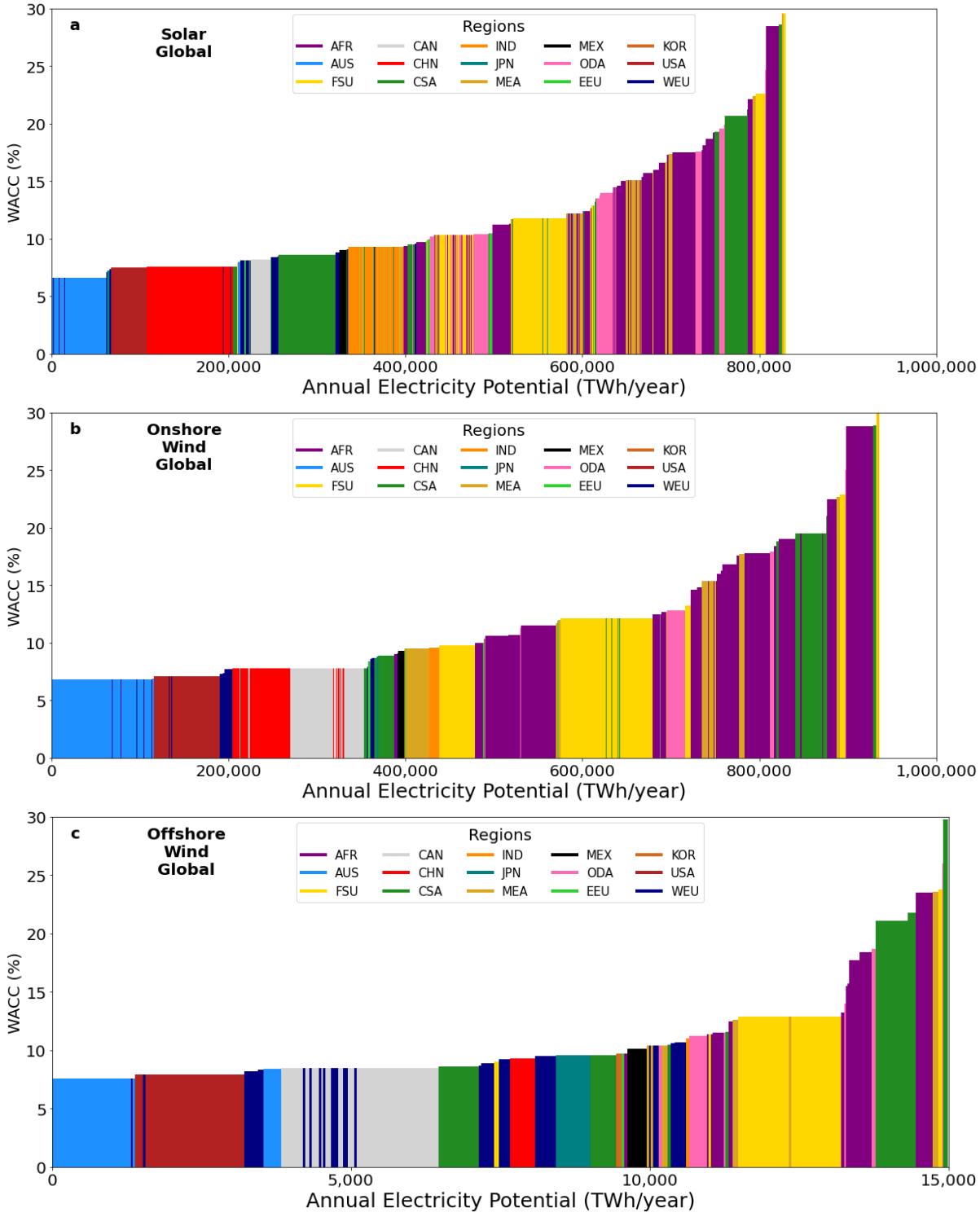
Extended Data Figures



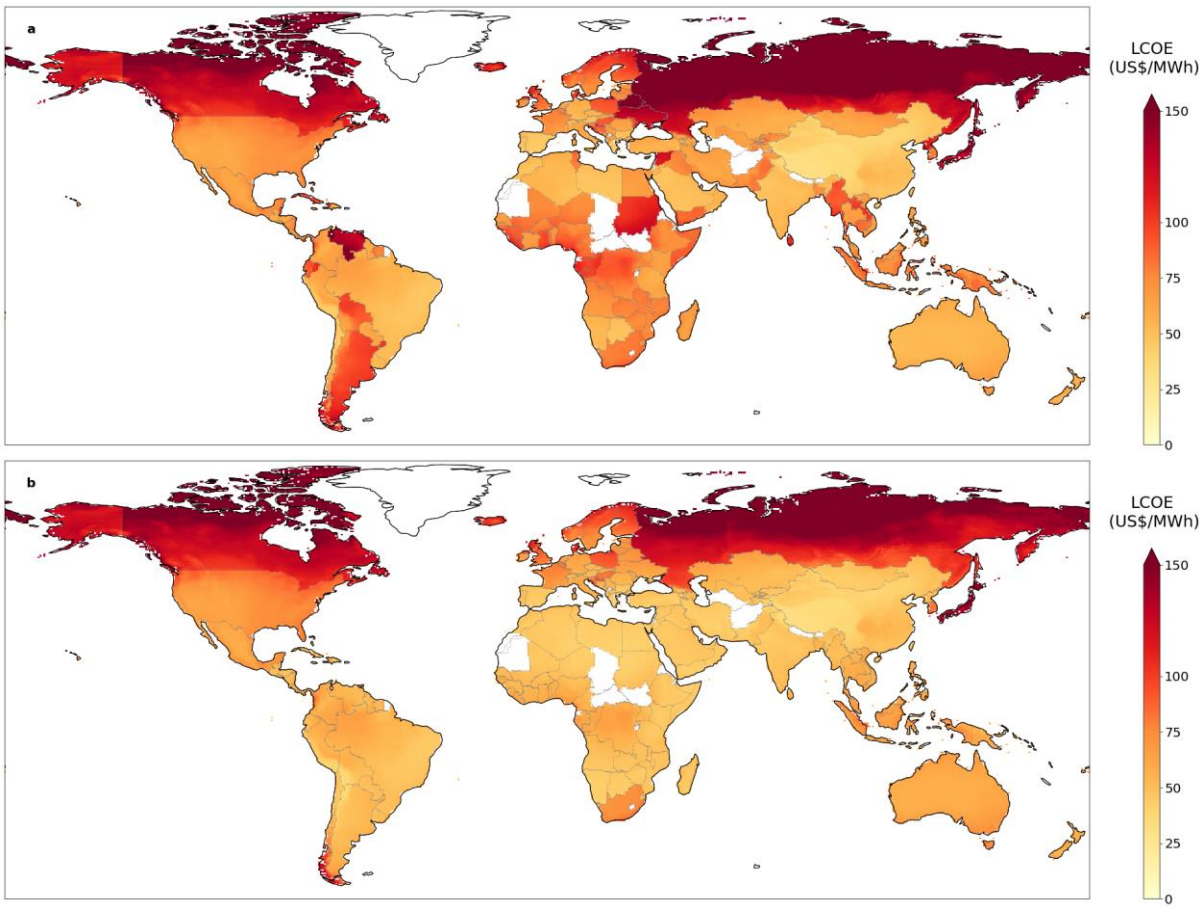
ED Figure 1: The annual average capacity factor for renewable energy generation. *Capacity factor is presented in % for a) solar PV and b) wind energy and are generated using the Renewables.Ninja method[61] and the PV Lib Python module developed by Sandia Labs[60]. The weather data used to generate wind and solar profiles is taken from NASA's MERRA-2 dataset[97].*



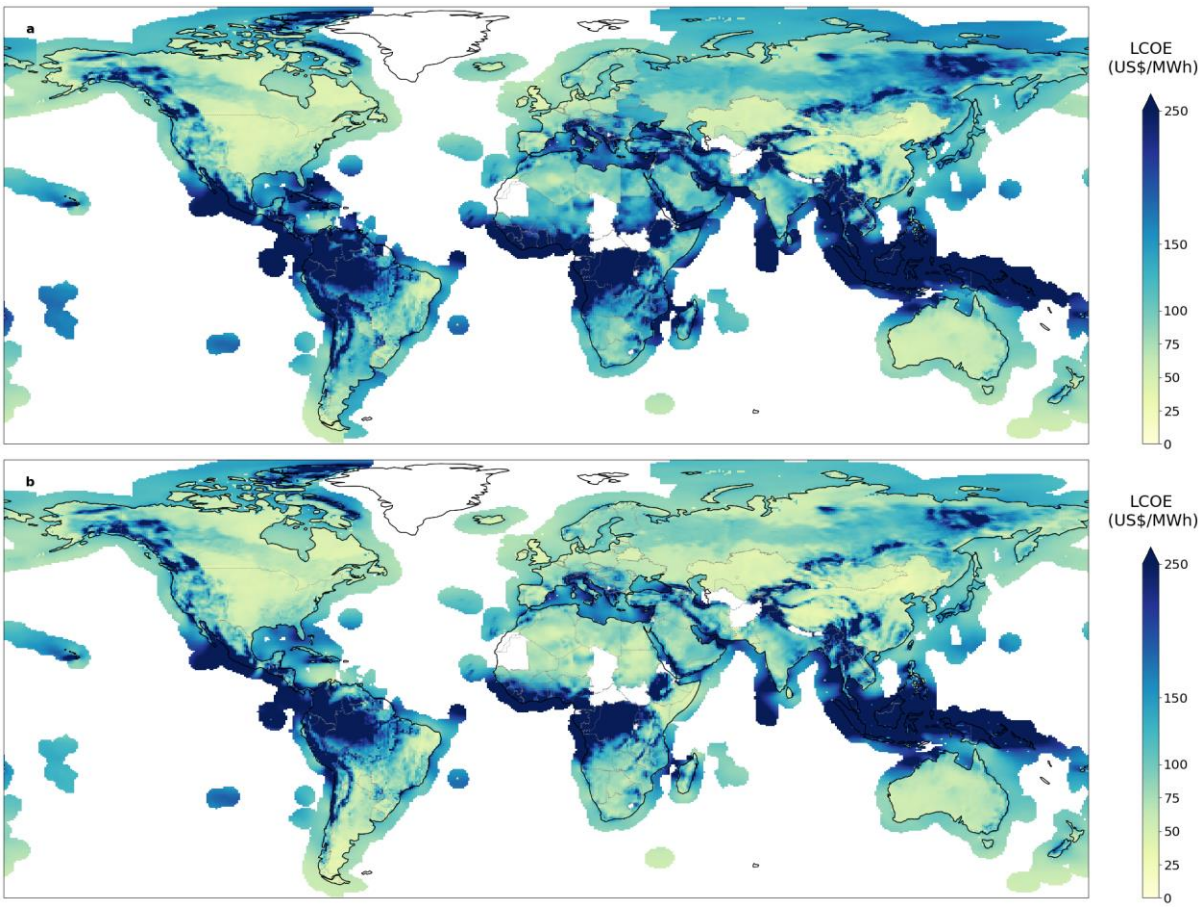
ED Figure 2: Estimates of the nominal weighted average cost of capital (WACC) for 2024, mapped globally, for a) Solar PV, and b) Wind power. *Results are shown on a country-level basis for the 176 countries that had data on country risk premiums available from Damodaran[30]. The WACC was estimated by disaggregating contributions to the cost of debt and equity, advancing on the approach developed by IRENA[54] (see Methods).*



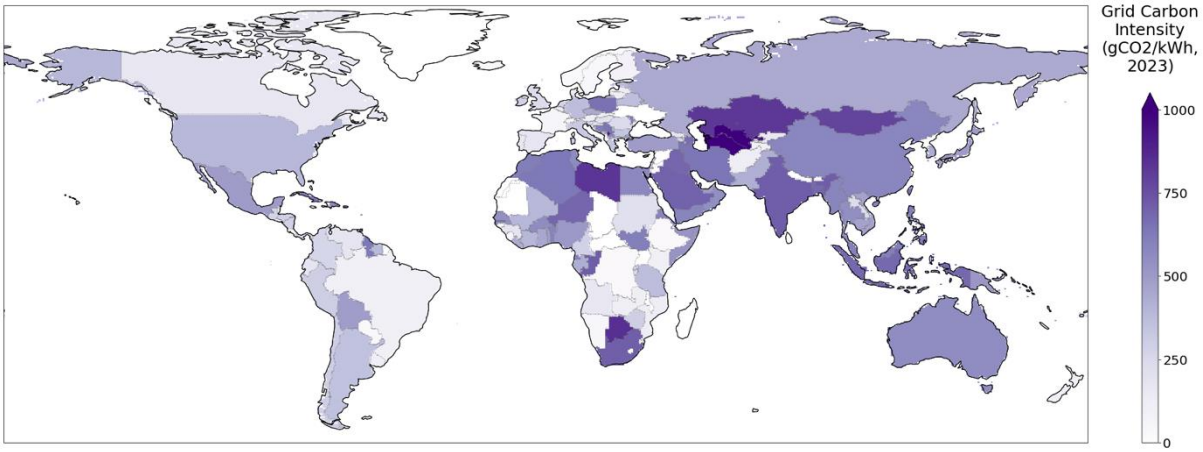
ED Figure 3: Global technical potential curves for a) solar PV, b) onshore wind and c) offshore wind, ordered by estimated technology and country-specific cost of capital for 2024 and with GDP per capita highlighted. *Note the different scale used for offshore wind electricity potential. Packing densities of 32.95, 6.52 and 4.0 MW per km^2 were used for solar, onshore wind and offshore wind, with land use constraints applied (see Supplementary). Minimum cut-offs of 10% and 18% average capacity factors across the year were applied to the solar and wind potentials to reflect constraints in economic feasibility[63]. AFR: Africa, AUS: Australia, FSU: Former Soviet Union, CAN: Canada, CHN: China, CSA: Central and South America, IND: India, JPN: Japan, MEA: Middle East, ODA: Other Developing Asia, EEU: Eastern Europe, KOR: Korea, USA: United States of America, WEU: Western Europe (see Supplementary for the full country-region mappings)*



ED Figure 4: Levelised cost of electricity from solar PV, plotted globally for a) country-specific costs of capital and b) a uniform nominal cost of capital of 8.2%. *The Uniform cost of capital was taken as the average estimated cost of capital for solar PV in the UNFCCC Annex II countries, which are commonly regarded as the developed country grouping (see Supplementary for the full list of Annex II countries).*



ED Figure 5: Levelised cost of electricity from onshore and offshore wind, plotted globally using a) country-specific costs of capital and b) a uniform nominal cost of capital of 8.1% and 9.0% for onshore and offshore, respectively. *The Uniform costs of capital are taken as the average estimated cost of capital for onshore and offshore wind in the UNFCCC Annex II countries, which are commonly regarded as the developed country grouping (see Supplementary for the full list of Annex II countries).*



ED Figure 6: Average grid electricity carbon intensity, plotted globally for 186 countries and/or regions. *Data is plotted in grams of carbon dioxide per kWh of generated electricity, taken from Ember Climate[47] for 2023; whilst more recent data for 2024 was available, this was only for 86 countries/regions globally.*

Methods

Renewable Power Profiles

Wind speed and solar irradiance data were sourced from NASA's MERRA-2[97], using a single representative year of data. Wind speeds were converted into hourly wind turbine power output using the Virtual Wind Farm (VWF) model[61] whilst solar PV power outputs were generated from the horizontal solar irradiance using Sandia Lab's PVLib package[60]. Two heuristics were used to specify the solar PV panel orientation: 1) PV modules were orientated so as to face the equator, and 2) the tilt angle was calculated using Equation [1], which was taken from the review conducted by Bamisile et al.[98]

$$\beta = 3\,|\varphi|^{\frac{2}{3}} \qquad [1]$$

where β is the tilt angle, and ϕ is the latitude, both in degrees.

Modelling the Cost of Capital

Building on the approach used by IRENA[54] and Calcaterra et al.[24], the cost of capital for solar PV, onshore wind and offshore wind was estimated at a country level for 2024 using national data on country risk and renewable penetration. The cost of capital differs for equity and debt financing components, with equity financing typically more expensive as payments to equity investors are made last in the event of project failure. Weighting the costs of debt and equity investment by their proportions, as in Equation [2], gives the total cost of capital for a project (referred to here as the WACC) for a country (x) and technology (t). C_E, C_D and R_D represent the cost of equity, cost of debt and debt share, respectively, with τ representing the corporate tax rate (as loan repayments are tax deductible in the majority of countries).

$$WACC^{x,t} = R_D^x C_D^{x,t}(1-\tau^x) + (1-R_D^x)C_E^{x,t} \qquad [2]$$

The cost of debt and equity were calculated individually at the country and technology level, for 2024, using an adaptation to the approach set out by IRENA[54] and used in Calcaterra et al.[24] with more detailed treatment of debt share in line with country risk. The cost of equity (C_e) for a given country (x) and technology (t) is calculated using Equation [3], with contributions from the risk-free rate ($r_{risk\text{-}free}$), country risk premium($r_{country}$), technology risk premium (r_{tech}) and the equity risk premium (r_{equity}). The latter accounts for the risk that equity investors take in being paid last in the event of project failure. The cost of debt (C_D) for a given country (x) and technology (t) is calculated using Equation [4], which includes the same risk-free rate and technology risks but include a country default spread (r_{cds}^x) and debt lenders margin (r_{debt}). The lenders margin is the profit that the lender aims to achieve when offering debt financing. Both the cost of debt and cost of equity were calculated for 2024.

$$C_E^{x,t} = r_{risk-free} + r_{country}^x + r_{tech}^{x,t} + r_{equity} \qquad [3]$$

$$C_D^{x,t} = r_{risk-free} + r_{cds}^x + r_{tech}^{x,t} + r_{debt} \qquad [4]$$

The risk-free rate was taken as the US Federal Reserve's interest rate on a 10 year government bond[99], as US Treasury Bonds are regarded as one of the safest investor assets and are typically used as a pricing benchmark. The country risk premium ($r^{x}_{country}$), country default spreads (r^{x}_{cds}) and equity risk premiums (r_{equity}) for 2024 were extracted from Damodaran's annual estimates of financial parameters[30], in line with Calcaterra et al.[24] and IRENA[54]. The debt lenders margin (r_{debt}) was taken as 2% in line with IRENA's assumptions for large-scale infrastructure projects[54]. Tax rates (τ) were taken from the Tax Foundations database of corporation tax rates[100].

The technology premium ($r^{x,t}_{tech}$) was evaluated in line with the maturity of each technology in a given country in 2024, calculated using annual generation capacity statistics taken from Ember Climate[47]. Countries were categorised into Mature, Intermediate and Emerging markets for each technology, with technology-specific premiums and boundaries for market maturity based on penetration. Mature markets for solar PV and onshore wind were categorised as over 10% penetration, whilst for offshore wind this was taken as 6%. For Intermediate markets, the lower boundary was 5% and 3% respectively, whilst all other countries were categorised as emerging markets in line with IRENA[54]. Technology premiums ranged from 1.5% to 3.25% for onshore wind and solar PV inversely with maturity, whilst for offshore wind the technology premium was assumed to range from 2.5% to 3.25%[54].

The debt share (R_D) was assumed to vary inversely between 40% and 80% in a linear fashion with country risk premium, with low-risk countries were assumed to have higher debt shares[54] whilst country risk premiums taken from Damodaran[30]. The boundaries of 40-80% were informed by the reported ranges in the Diacore project for the EU-28, which ranged between 55-80% for solar PV and onshore wind[101], extended to account for the lower country risk premiums in the EU compared to other countries and regions in the world[30]. Notably, this improves on the approach used by IRENA[54] which models a debt share of either 60%, 70% or 80% depending on the maturity of the technology in each market (without accounting for country-level risks).

Levelised Cost of Electricity

The levelised cost of electricity (LCOE) is a common metric used to compare the economic viability of producing electricity at different locations and/or from different sources[11]. It represents the value at which electricity would need to be sold at to give a net present value of zero. Equation [6] was used to calculate the LCOE, which requires input values for the capital expenditure (CAPEX), annual operating expenditure (OPEX) and electricity production (E_t) for each year (t). The costs and electricity production in each year are then discounted using a discount rate (r), up to the plant's assumed economic lifetime of 20 years (n). The country- and technology-specific WACC values estimated by this work using Equation [5] were used for the discount rate, reflecting the estimated financing terms for renewable projects.

$$LCOE = \frac{\sum_{t}^{n} \frac{CAPEX_t + OPEX_t}{(1+r)^t}}{\sum_{t}^{n} \frac{E_t}{(1+r)^t}} \qquad [6]$$

Capital costs for solar PV and onshore wind were taken from IRENA's Renewable Generation Costs 2023 and applied at a regional level[11], with benchmarking also conducted against the GNESTE database[94]. Capital costs for onshore wind are available at a regional level, whilst capital costs for solar PV are provided by IRENA for the 30 largest markets. The weighted average for solar PV for each region was calculated and used for countries in regions without cost data, whilst for regions without any cost data the global weighted average cost of US$876/kW was used[11] (see **Supplementary** for full details at a country level). Annual operating costs for solar PV, onshore and offshore wind were taken as 2.3%, 2.6% and 3% of the initial capital investment, respectively, using the same assumptions used by the International Energy Agency[102].

Technical Potential

To generate national and regional LCOE-supply curves, the technical renewable potential at each gridpoint of resolution 0.5 latitude by 0.625 longitude modelled in the MERRA-2 meteorological data[97] was evaluated in this work. Packing densities for solar PV was taken to be 32.95 MW/km^2, based on NREL's estimates for fixed-axis utility scale generation[103]. For onshore wind, a packing density of 6.52 MW/km^2 was taken from Bosch[49]. Finally, for offshore wind, a packing density of 4 MW/km^2 was extracted from NREL's evaluation of capacity density considerations for offshore wind[104].

The production potential at each location also depends on the available land that is suitable for renewable power, which may be affected by economic, technical or environmental constraints. Land use data was extracted using the GlobCover dataset[105], which is generated from satellite data and categorised land use into 23 classes at a high spatial resolution. Corresponding land availability factors for wind and solar were extracted from Dupont 2020[106] and Dupont 2018[107] (see **Supplementary**) and used as a constraint to the annual production potential and maximum installed capacity at each location. Land use availability factors of 100% were assumed for offshore locations.

Renewable abatement potential

Assessing lifetime renewable abatement potential requires making assumptions around the trajectory of future grid intensities at a national level. Using Equation [7], the total abatement over the project lifetime (n) was calculated using the annual technical electricity potential (E_t) at each grid cell and the national grid carbon intensity (GI_t), based on an assumption that each MWh of electricity generated from the solar or wind farm displaces a MWh of existing grid electricity. Initial grid intensities for the first year were taken from Ember Climate[47] for 2023. Grid intensity was modelled as falling linearly to zero over the 20-year project lifetime, i.e. that global average grid intensity reaches zero by 2045 (in line with the Net Zero Scenario from the International Energy Agency[4]). It is the case both in advanced economies and developing economies, though advanced economies are modelled to reach zero grid emissions intensities by approximately 2035 under the Net Zero Emissions scenario[4]. Lifetime carbon abatement potential estimates may be lower therefore in developed countries.

$$Abatement = \sum_{t}^{n} E_t * GI_t \quad [7]$$

To estimate total abatement potential for each country, data on total annual electricity generation in 2023 was also sourced from Ember Climate[47]. The technical potential (E_t) for each region was limited to the total annual generation, with the lowest cost locations used to create the cost-abatement supply curves. To calculate the abatement potential at each grid cell, 10% of the technical potential at each grid cell was assumed to be used to support the decarbonisation of national power grids, reflecting conflicting demands for land and other limits to infrastructure deployment e.g., grid connections.

Data Availability

Data supporting the findings of this study is either included in the paper or is available upon request.

Code Availability

The Python scripts for data processing and visualisation can be accessed in the GitHub repository, found at https://github.com/LukeHatton21/CostofCapitalLCOEModel

Competing Interests

The authors declare no conflicting interests.

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Author CRediT Statement

Luke Hatton: Conceptualization, Methodology, Formal analysis, Data curation, Visualisation, Writing – Original draft preparation. **Gbemi Oluleye:** Funding acquisition, Supervision, Writing – Review & Editing. **Malte Jansen:** Funding acquisition, Writing – Review & Editing. **Iain Staffell:** Funding acquisition, Writing – Review & Editing. **Adam Hawkes:** Funding acquisition, Supervision, Writing – Review & Editing.

References

1. IRENA. *Key Enablers to Triple Renewables by 2030: Finance*. https://www.irena.org/News/articles/2024/Aug/Key-Enablers-to-Triple-Renewables-by-2030-Finance.
2. UNFCCC & COP28 Presidency. *Summary of Global Climate Action at COP28*. https://unfccc.int/sites/default/files/resource/Summary_GCA_COP28.pdf (2023).
3. BloombergNEF. *Energy Transition Investment Trends 2024*. https://about.bnef.com/energy-transition-investment.
4. IEA. *World Energy Outlook 2024*. https://www.iea.org/reports/world-energy-outlook-2024 (2024).
5. Energy Institute, K. *Energy Institute 2024 Statistical Review of World Energy*. https://www.energyinst.org/statistical-review (2024).
6. IRENA. *World Energy Transitions Outlook 2023: 1.5°C Pathway*. https://www.irena.org/Publications/2023/Jun/World-Energy-Transitions-Outlook-2023.
7. UNFCCC. COP28 Agreement Signals "Beginning of the End" of the Fossil Fuel Era. (2023).
8. Way, R., Ives, M. C., Mealy, P. & Farmer, J. D. Empirically grounded technology forecasts and the energy transition. *Joule* **6**, 2057–2082 (2022).
9. Xiao, M., Junne, T., Haas, J. & Klein, M. Plummeting costs of renewables - Are energy scenarios lagging? *Energy Strategy Reviews* **35**, 100636 (2021).
10. Lazard. *Levelized Cost of Energy and Levelized Cost of Storage – 2024*. https://www.lazard.com/research-insights/levelized-cost-of-energyplus/ (2024).
11. IRENA. Renewable Power Generation Costs in 2023. (2024).
12. IEA. *COP28 Tripling Renewable Capacity Pledge: Tracking Countries' Ambitions and Identifying Policies to Bridge the Gap*. https://www.iea.org/reports/cop28-tripling-renewable-capacity-pledge (2024).
13. *Tripling Renewable Power and Doubling Energy Efficiency by 2030 Crucial Steps towards 1.5°C*. (International Renewable Energy Agency, Abu Dhabi, 2023).
14. Shields, M., Beiter, P., Nunemaker, J., Cooperman, A. & Duffy, P. Impacts of turbine and plant upsizing on the levelized cost of energy for offshore wind. *Applied Energy* **298**, 117189 (2021).
15. Kempa, K., Moslener, U. & Schenker, O. The cost of debt of renewable and non-renewable energy firms. *Nat Energy* **6**, 135–142 (2021).
16. Egli, F., Steffen, B. & Schmidt, T. S. A dynamic analysis of financing conditions for renewable energy technologies. *Nat Energy* **3**, 1084–1092 (2018).
17. Jung, C. & Schindler, D. Reasons for the Recent Onshore Wind Capacity Factor Increase. *Energies* **16**, 5390 (2023).
18. Christensen, L. OPEX Benchmark – An insight into the operational expenditures of European offshore wind farms. (2022).
19. IEA. *World Energy Employment 2023*. 27 https://iea.blob.core.windows.net/assets/ba1eab3e-8e4c-490c-9983-80601fa9d736/World_Energy_Employment_2023.pdf.

20. Schmidt, T. S., Born, R. & Schneider, M. Assessing the costs of photovoltaic and wind power in six developing countries. *Nature Clim Change* **2**, 548–553 (2012).
21. Seetharaman, Moorthy, K., Patwa, N., Saravanan & Gupta, Y. Breaking barriers in deployment of renewable energy. *Heliyon* **5**, e01166 (2019).
22. RISE. RISE: Regulatory Indicators for Sustainable Energy. *RISE* https://rise.esmap.org/analytics (2024).
23. Afful-Dadzie, A. Global 100% energy transition by 2050: A fiction in developing economies? *Joule* **5**, 1641–1643 (2021).
24. Calcaterra, M. *et al.* Reducing the cost of capital to finance the energy transition in developing countries. *Nat Energy* (2024) doi:10.1038/s41560-024-01606-7.
25. Briera, T. & Lefèvre, J. Reducing the cost of capital through international climate finance to accelerate the renewable energy transition in developing countries. *Energy Policy* **188**, 114104 (2024).
26. Buchner, B. *et al. Global Landscape of Climate Finance 2023*. https://www.climatepolicyinitiative.org/publication/global-landscape-of-climate-finance-2023/ (2023).
27. Cornieti, S. & Nicolas, C. How to Unlock Pipelines of Bankable Renewable Energy Projects in Emerging Markets and Developing Countries? (2023).
28. Mazzucato, M. & Semieniuk, G. Financing renewable energy: Who is financing what and why it matters. *Technological Forecasting and Social Change* **127**, 8–22 (2018).
29. Gatzert, N. & Vogl, N. Evaluating investments in renewable energy under policy risks. *Energy Policy* **95**, 238–252 (2016).
30. Aswath Damodaran. Country Default Spreads and Risk Premiums. https://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/ctryprem.html (2024).
31. IEA. *Reducing the Cost of Capital: Strategies to Unlock Clean Energy Investment in Emerging and Developing Economies*. https://www.iea.org/reports/reducing-the-cost-of-capital (2024).
32. Hassan, T. A. & Zhang, T. The Economics of Currency Risk. *Annu. Rev. Econ.* **13**, 281–307 (2021).
33. Kwabi, F., Owusu, A., Ezeani, E. & Boateng, A. The impact of political uncertainty on the cost of capital. *Rev Quant Finan Acc* **62**, 1397–1429 (2024).
34. Damodaran, A. Country Risk: Determinants, Measures and Implications - The 2022 Edition. *SSRN Journal* (2022) doi:10.2139/ssrn.4161010.
35. Ameli, N. *et al.* Higher cost of finance exacerbates a climate investment trap in developing economies. *Nat Commun* **12**, 4046 (2021).
36. Polzin, F. *et al.* The effect of differentiating costs of capital by country and technology on the European energy transition. *Climatic Change* **167**, 26 (2021).
37. Bullard, N. Decarbonization: Stocks and flows, abundance and scarcity, net zero. (2024).
38. Vartiainen, E., Masson, G., Breyer, C., Moser, D. & Román Medina, E. Impact of weighted average cost of capital, capital expenditure, and other parameters on future utility-scale PV levelised cost of electricity. *Prog Photovolt Res Appl* **28**, 439–453 (2020).
39. Egli, F., Steffen, B. & Schmidt, T. S. Bias in energy system models with uniform cost of capital assumption. *Nat Commun* **10**, 4588 (2019).
40. Schmidt, T. S. *et al.* Adverse effects of rising interest rates on sustainable energy transitions. *Nat Sustain* **2**, 879–885 (2019).

41. Adelaja, A. O. Barriers to national renewable energy policy adoption: Insights from a case study of Nigeria. *Energy Strategy Reviews* **30**, 100519 (2020).
42. Scott, S. The grant element method of measuring the concessionality of loans and debt relief. (2017).
43. International Energy Agency & International Finance Corporation. *Scaling up Private Finance for Clean Energy in Emerging and Developing Economies*. https://www.iea.org/reports/scaling-up-private-finance-for-clean-energy-in-emerging-and-developing-economies (2023).
44. Bhattacharya, A., Songwe, V., Soubeyran, E. & Stern, N. *Raising Ambition and Accelerating Delivery of Climate Finance*. https://www.lse.ac.uk/granthaminstitute/wp-content/uploads/2024/11/Raising-ambition-and-accelerating-delivery-of-climate-finance_Third-IHLEG-report.pdf (2024).
45. Sieber, A. & Erzini Vernoit, I. Quantifying international public finance provision needs for the new UN climate finance goal. *npj Clim. Action* **3**, 104 (2024).
46. Ondraczek, J., Komendantova, N. & Patt, A. WACC the dog: The effect of financing costs on the levelized cost of solar PV power. *Renewable Energy* **75**, 888–898 (2015).
47. Ember Climate. Yearly Electricity Data. https://ember-energy.org/data/yearly-electricity-data/.
48. Prăvălie, R., Patriche, C. & Bandoc, G. Spatial assessment of solar energy potential at global scale. A geographical approach. *Journal of Cleaner Production* **209**, 692–721 (2019).
49. Bosch, J., Staffell, I. & Hawkes, A. D. Temporally-explicit and spatially-resolved global onshore wind energy potentials. *Energy* **131**, 207–217 (2017).
50. Kerr, S. & Hu, X. Filling the climate finance gap: holistic approaches to mobilise private finance in developing economies. *npj Clim. Action* **4**, 16 (2025).
51. Bolson, N., Prieto, P. & Patzek, T. Capacity factors for electrical power generation from renewable and nonrenewable sources. *Proc. Natl. Acad. Sci. U.S.A.* **119**, e2205429119 (2022).
52. Holtinen, H. *et al.* Estimating the Reduction of Generating System CO2 Emissions Resulting from Significant Wind Energy Penetration.
53. Steffen, B. The importance of project finance for renewable energy projects. *Energy Economics* **69**, 280–294 (2018).
54. IRENA. *The Cost of Financing for Renewable Power*. https://www.irena.org/Publications/2023/May/The-cost-of-financing-for-renewable-power (2023).
55. Egli, F., Orgland, N., Taylor, M., Schmidt, T. S. & Steffen, B. Estimating the cost of capital for solar PV projects using auction results. *Energy Policy* **183**, 113849 (2023).
56. Steffen, B. Estimating the cost of capital for renewable energy projects. *Energy Economics* **88**, 104783 (2020).
57. Lonergan, K. E. *et al.* Improving the representation of cost of capital in energy system models. *Joule* **7**, 469–483 (2023).
58. Agutu, C., Egli, F., Williams, N. J., Schmidt, T. S. & Steffen, B. Accounting for finance in electrification models for sub-Saharan Africa. *Nat Energy* **7**, 631–641 (2022).
59. Fisch-Romito, V., Jaxa-Rozen, M., Wen, X. & Trutnevyte, E. Multi-country evidence on societal factors to include in energy transition modelling. *Nat Energy* **10**, 460–469 (2025).
60. F. Holmgren, W., W. Hansen, C. & A. Mikofski, M. pvlib python: a python package for modeling solar energy systems. *JOSS* **3**, 884 (2018).

61. Staffell, I. & Pfenninger, S. Using bias-corrected reanalysis to simulate current and future wind power output. *Energy* **114**, 1224–1239 (2016).
62. IMF. *World Economic Outlook 2025*. https://www.imf.org/en/Publications/WEO/Issues/2025/04/22/world-economic-outlook-april-2025 (2025).
63. Chu, C.-T. & Hawkes, A. D. A geographic information system-based global variable renewable potential assessment using spatially resolved simulation. *Energy* **193**, 116630 (2020).
64. Alova, G., Trotter, P. A. & Money, A. A machine-learning approach to predicting Africa's electricity mix based on planned power plants and their chances of success. *Nat Energy* **6**, 158–166 (2021).
65. IEA & IAEA. Projected Costs of Generating Electricity. (2020).
66. Maclaurin, G. *et al. The Renewable Energy Potential (reV) Model: A Geospatial Platform for Technical Potential and Supply Curve Modeling*. NREL/TP--6A20-73067, 1563140, MainId:13369 https://www.osti.gov/servlets/purl/1563140/ (2021) doi:10.2172/1563140.
67. Anandarajah, G., Pye, S., Usher, W., Kesicki, F. & Mcglade, C. *TIAM-UCL Global Model Documentation: Working Paper*. https://www.researchgate.net/profile/Gabrial-Anandarajah/publication/239917973_TIAM-UCL_Global_Model_Documentation/links/553a6def0cf29b5ee4b6354e/TIAM-UCL-Global-Model-Documentation.pdf (2011).
68. Jansen, M. *et al.* Offshore wind competitiveness in mature markets without subsidy. *Nat Energy* **5**, 614–622 (2020).
69. Global Energy Monitor. Global Wind Power Tracker. (2024).
70. McCoy, A. *et al. Offshore Wind Market Report: 2024 Edition*. https://www.nrel.gov/docs/fy24osti/90525.pdf (2024).
71. Ember Climate. Electricity Data Explorer. *Ember Climate* https://ember-energy.org/data/electricity-data-explorer/ (2025).
72. Sweerts, B., Longa, F. D. & Van Der Zwaan, B. Financial de-risking to unlock Africa's renewable energy potential. *Renewable and Sustainable Energy Reviews* **102**, 75–82 (2019).
73. Navia Simon, D. & Diaz Anadon, L. Power price stability and the insurance value of renewable technologies. *Nat Energy* **10**, 329–341 (2025).
74. Reuters. Bangladesh halts expensive spot LNG imports despite load-shedding. *Reuters* (2022).
75. Lolla, Aditya. Spot market LNG purchases can cost Bangladesh about $11 billion between 2022 & 2024. *Ember* https://ember-energy.org/latest-insights/spot-market-lng-purchases-can-cost-bangladesh-about-11-billion-between-2022-2024/ (2022).
76. Garcia-Casals, X., Ferroukhi, R. & Parajuli, B. Measuring the socio-economic footprint of the energy transition. *Energy Transit* **3**, 105–118 (2019).
77. Carley, S. & Konisky, D. M. The justice and equity implications of the clean energy transition. *Nat Energy* **5**, 569–577 (2020).
78. Semieniuk, G. *et al.* Stranded fossil-fuel assets translate to major losses for investors in advanced economies. *Nat. Clim. Chang.* **12**, 532–538 (2022).
79. Schmidt, T. S. Low-carbon investment risks and de-risking. *Nature Clim Change* **4**, 237–239 (2014).

80. International Energy Agency. Cost of Capital Observatory. https://www.iea.org/reports/cost-of-capital-observatory/dashboard-2 (2023).
81. Bhandary, R. R., Gallagher, K. S. & Zhang, F. Climate finance policy in practice: a review of the evidence. *Climate Policy* **21**, 529–545 (2021).
82. International Energy Agency. *Roadmap to Increase Investment in Clean Energy in Developing Countries – an Initiative by the G20 Brazil Presidency*. https://iea.blob.core.windows.net/assets/6ac243a9-247b-4b79-bc01-0e7730434118/RoadmaptoIncreaseInvestmentinCleanEnergyinDevelopingCountriesaninitiativebytheG20BrazilPresidency.pdf (2024).
83. Bhattacharya, A., Songwe, V., Soubeyran, E. & Stern, N. A climate finance framework: decisive action to deliver on the Paris Agreement. (2023).
84. The World Bank. What You Need to Know About Concessional Finance for Climate Action. https://www.worldbank.org/en/news/feature/2021/09/16/what-you-need-to-know-about-concessional-finance-for-climate-action (2021).
85. UNFCCC & COP29 Presidency. New collective quantified goal on climate finance.
86. Climate Action Network. Betrayal in Baku: developed countries fail people and planet. *Betrayal in Baku: developed countries fail people and planet* https://climatenetwork.org/2024/11/23/cop29_betrayal_in_baku/ (2024).
87. Arezki, R. Climate finance for Africa requires overcoming bottlenecks in domestic capacity. *Nat. Clim. Chang.* **11**, 888–888 (2021).
88. Özdemir, Ö., Hobbs, B. F., Van Hout, M. & Koutstaal, P. R. Capacity vs energy subsidies for promoting renewable investment: Benefits and costs for the EU power market. *Energy Policy* **137**, 111166 (2020).
89. Luscombe, H. *et al.* Data-to-Deal (D2D): An Emerging and Effective Approach to Financing the Climate Transition. Preprint at https://doi.org/10.33774/coe-2024-21xv4 (2024).
90. Nijsse, F. J. M. M. *et al.* The momentum of the solar energy transition. *Nat Commun* **14**, 6542 (2023).
91. Đukan, M. & Kitzing, L. A bigger bang for the buck: The impact of risk reduction on renewable energy support payments in Europe. *Energy Policy* **173**, 113395 (2023).
92. Ordonez, J. A., Vandyck, T., Keramidas, K., Garaffa, R. & Weitzel, M. Just Energy Transition Partnerships and the future of coal. *Nat. Clim. Chang.* **14**, 1026–1029 (2024).
93. Pfenninger, S., Hawkes, A. & Keirstead, J. Energy systems modeling for twenty-first century energy challenges. *Renewable and Sustainable Energy Reviews* **33**, 74–86 (2014).
94. Hatton, L. *et al.* The global and national energy systems techno-economic (GNESTE) database: Cost and performance data for electricity generation and storage technologies. *Data in Brief* 110669 (2024) doi:10.1016/j.dib.2024.110669.
95. Fuchs, J. L., Tesfamichael, M., Clube, R. & Tomei, J. How does energy modelling influence policymaking? Insights from low- and middle-income countries. *Renewable and Sustainable Energy Reviews* **203**, 114726 (2024).
96. Đukan, M. & Steffen, B. Cost of capital for renewables and enabling technologies: Measuring the multidimensional heterogeneity in Switzerland. *Applied Energy* **390**, 125822 (2025).
97. Gelaro, R. *et al.* The Modern-Era Retrospective Analysis for Research and Applications, Version 2 (MERRA-2). *J. Climate* **30**, 5419–5454 (2017).
98. Bamisile, O., Acen, C., Cai, D., Huang, Q. & Staffell, I. The environmental factors affecting solar photovoltaic output. *Renewable and Sustainable Energy Reviews* **208**, 115073 (2025).

99. OECD. Long-term interest rates. doi:10.1787/662d712c-en.
100. Enache, C. Corporate Tax Rates Around the World, 2024. *Tax Foundation* https://taxfoundation.org/data/all/global/corporate-tax-rates-by-country-2024/.
101. DiaCore, Fraunhofer ISI, ECOFYS. *The Impact of Risks in Renewable Energy Investments and the Role of Smart Policies*. https://www.isi.fraunhofer.de/content/dam/isi/dokumente/ccx/dia-core/D3-4_diacore_2016_impact_of_risk_in_res_investments.pdf (2016).
102. International Energy Agency. Levelised Cost of Hydrogen Maps. https://www.iea.org/data-and-statistics/data-tools/levelised-cost-of-hydrogen-maps (2023).
103. NREL. US Solar Photovoltaic System and Energy Storage Cost Benchmarks, 2023.
104. Mulas Hernando, D., Musial, W., Duffy, P. & Shields, M. *Capacity Density Considerations for Offshore Wind Plants in the United States*. https://www.nrel.gov/docs/fy24osti/86933.pdf (2023).
105. Bontemps, S. *et al.* GLOBCOVER 2009. (2011).
106. Dupont, E., Koppelaar, R. & Jeanmart, H. Global available solar energy under physical and energy return on investment constraints. *Applied Energy* **257**, 113968 (2020).
107. Dupont, E., Koppelaar, R. & Jeanmart, H. Global available wind energy with physical and energy return on investment constraints. *Applied Energy* **209**, 322–338 (2018).

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